PART II: PATTERNS

Topic 18: Year-over-Year (YoY) Comparison

The Core Concept

Year-over-Year (YoY) comparison means comparing a given period to the **same period in the prior year**. For example, comparing July 2023 to July 2022 — not to June 2023.

Why It Matters

This is one of the most practical tools in public safety data analysis. "How are we doing compared to last month?" is often the wrong question — it mixes up seasonal fluctuation with real change. "How are we doing compared to the same month last year?" **controls for seasonality** and reveals genuine progress or deterioration.

The question every commander, mayor, and police chief actually needs to answer: "Are we better off than we were a year ago at this same point?"

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	2021 (N. North)	2022 (N. North)	2023 (N. North)	YoY Change (2022→2023)	Signal
Jan	28	22	12	−45%	✅ Improving
Feb	26	20	10	−50%	✅ Improving
Mar	30	24	14	−42%	✅ Improving
Jun	40	32	22	−31%	✅ Improving
Jul	42	35	25	−29%	✅ Improving
Oct	30	24	16	−33%	✅ Improving
Dec	25	18	11	−39%	✅ Improving

Every single month in 2023 shows improvement compared to 2022 — and dramatic improvement compared to 2021. The YoY comparison makes this unambiguous, even during months that might look "high" compared to a January baseline.

Without YoY: Seeing July at 25 might cause concern. **With YoY:** July 25 (2023) vs. July 35 (2022) = a 29% reduction. That is a success, not a crisis.

💡 **The Insight:** Year-over-Year comparison is the simplest way to remove seasonal noise and show genuine directional change.

Key Takeaway: When presenting performance data to leadership, always include a YoY column. It frames the current month in the context that actually matters.